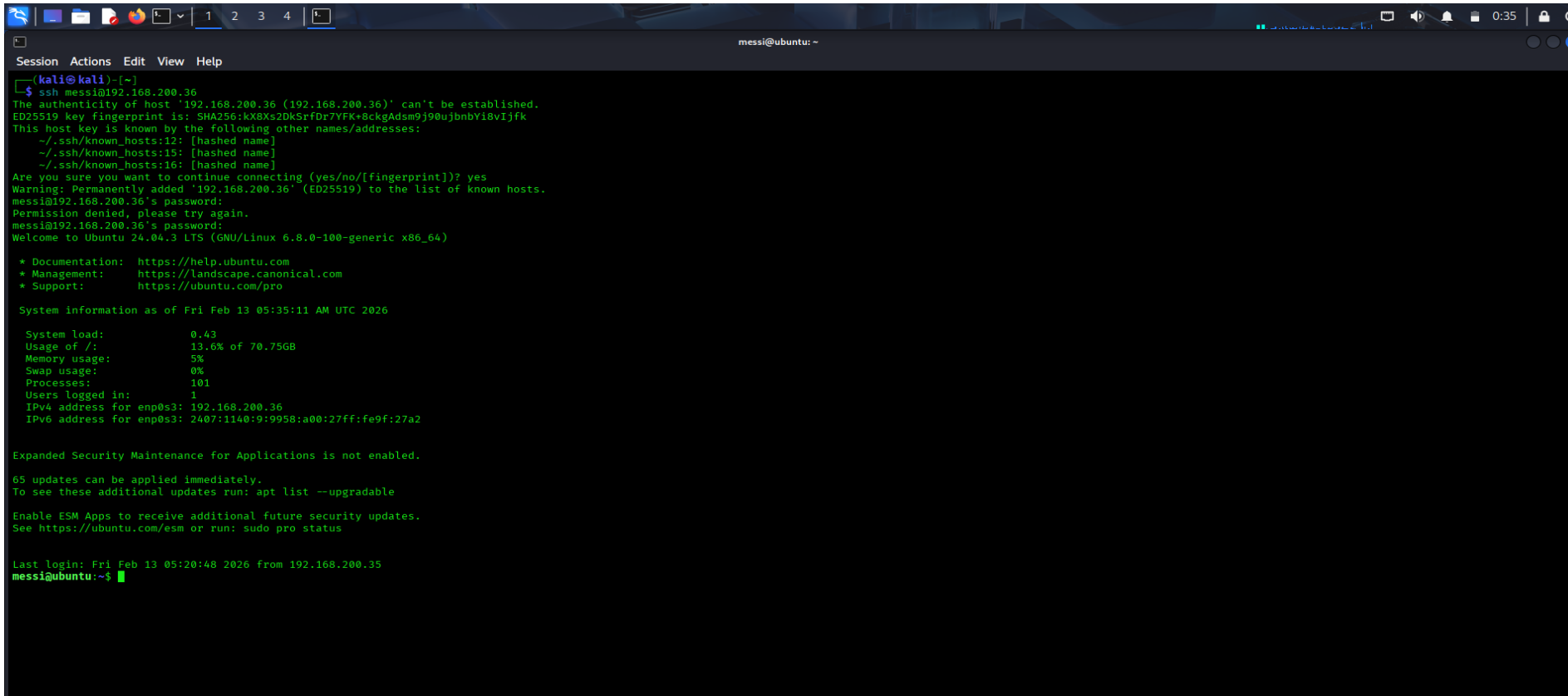


Monitoring Failed SSH Login Activities in Splunk

NAME : DITTO.D

Step 1 : Connect SSH in the kali



```
(kali@kali)~$ ssh messi@192.168.200.36
The authenticity of host '192.168.200.36 (192.168.200.36)' can't be established.
ED25519 key fingerprint is: SHA256:kX8Xs2DkSrFDr7YFK+8ckgAdsm9j90ujbny18vIjfk
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:12: [hashed name]
  ~/.ssh/known_hosts:15: [hashed name]
  ~/.ssh/known_hosts:16: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.200.36' (ED25519) to the list of known hosts.
messi@192.168.200.36's password:
Permission denied, please try again.
messi@192.168.200.36's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Feb 13 05:35:11 AM UTC 2026

System load:          0.43
Usage of /:            13.6% of 70.75GB
Memory usage:         5%
Swap usage:           0%
Processes:            101
Users logged in:      1
IPv4 address for enp0s3: 192.168.200.36
IPv6 address for enp0s3: 2407:1140:9:9958:a00:27ff:fe9f:27a2

Expanded Security Maintenance for Applications is not enabled.

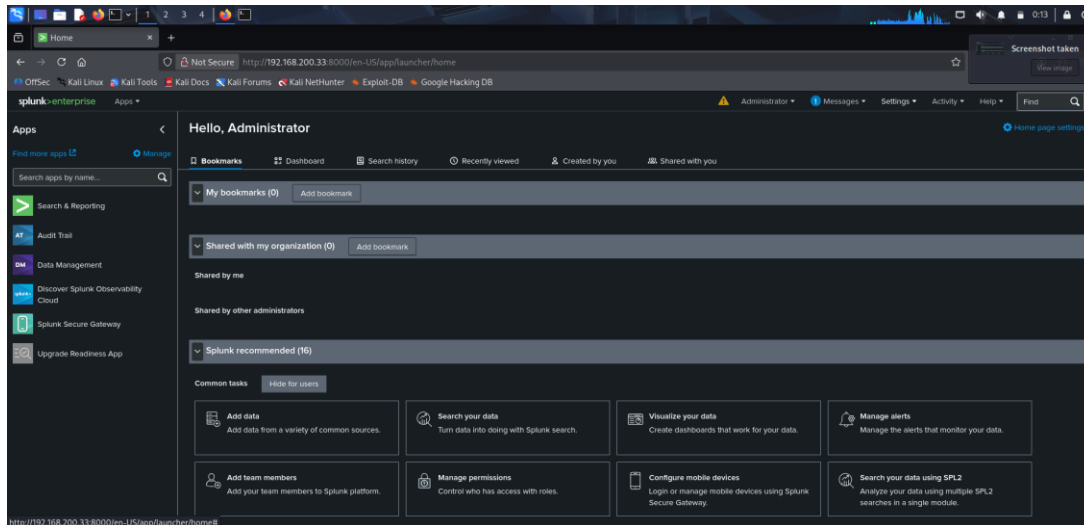
65 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Feb 13 05:20:48 2026 from 192.168.200.35
messi@ubuntu:~$
```

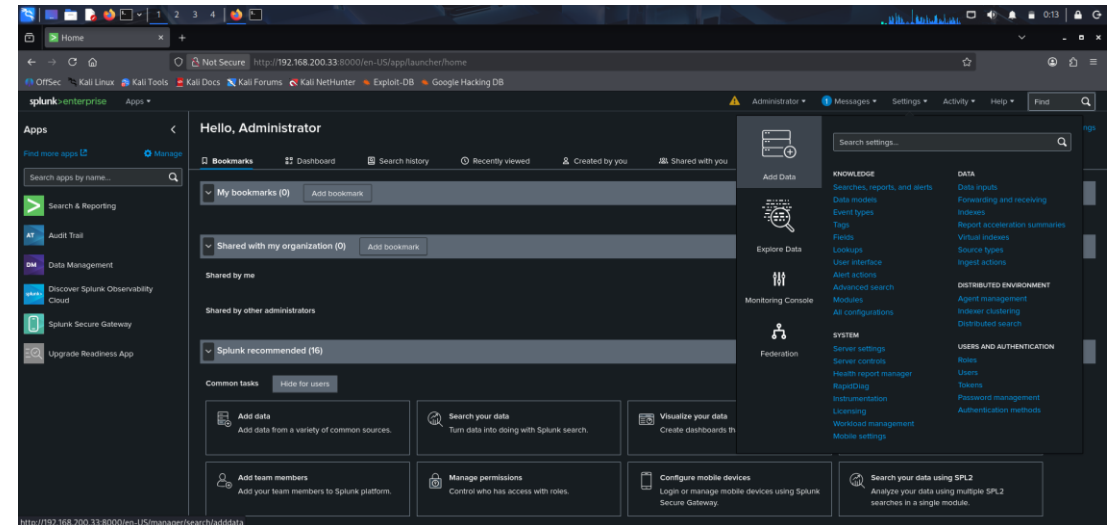
If you give wrong password here that was fine. It's used for after some time.

Step 2 : Login Splunk :



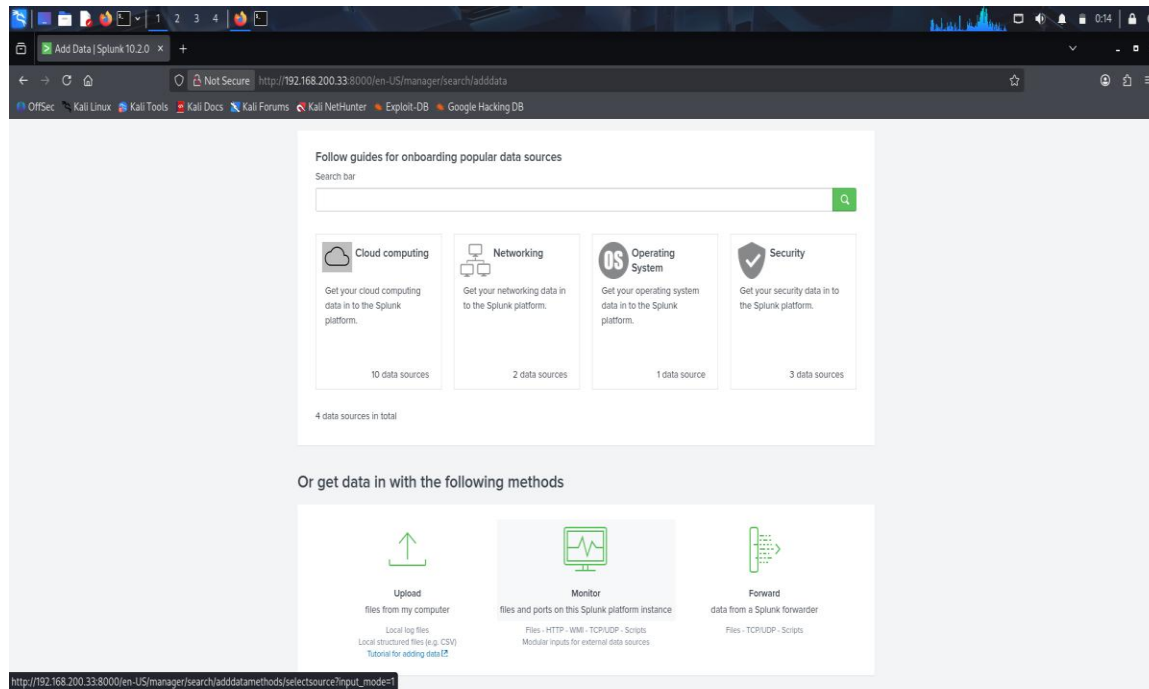
We need to login splunk

Step 3 : Make sure Ubuntu SSH Logs are Ingested into splunk :

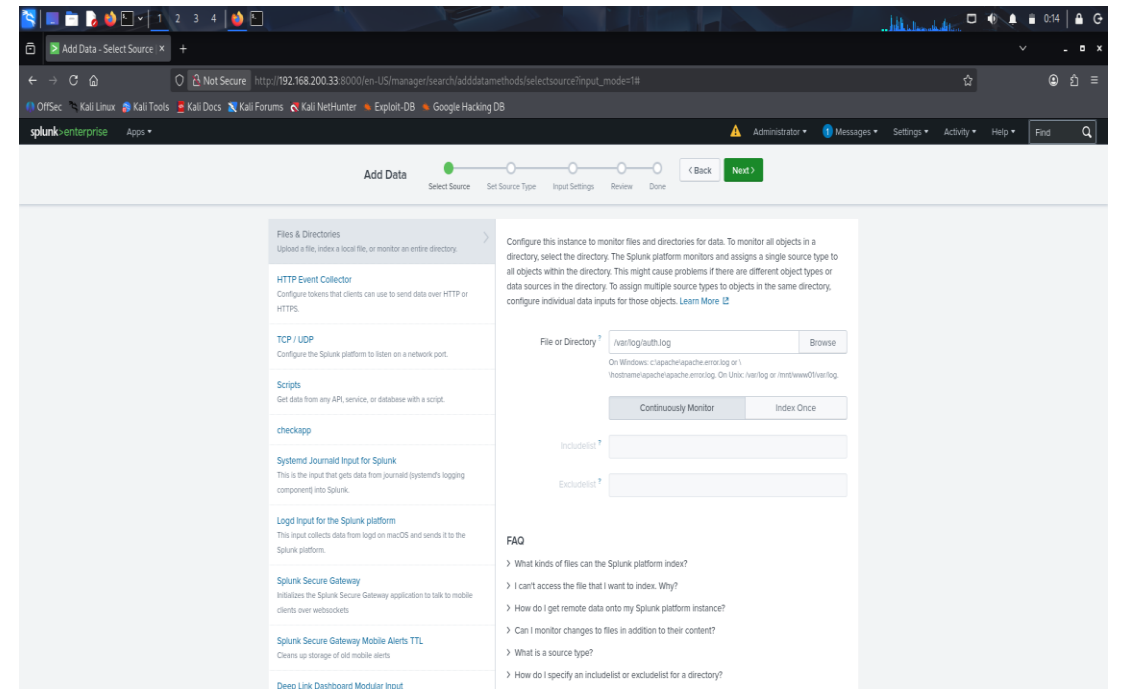


Go to settings – Click – Add Data

Click – Monitor



Files & Directories : var/log/auth.log



Change – Source type : Operating system – Linux secure

Click – Review

Add Data - Set Source Type

Not Secure http://192.168.200.33:8000/en-US/manager/search/adddatamethods/datapreview

Administrator Messages Settings Activity Help Find

Add Data Select Source Set Source Type Input Settings Review Done Back Next

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: `/var/log/auth.log`

Source type: linux_secure Save As

Format Select... Select... Prev 1 2 3 4 5 6 7 8 Next

	Time	Event
1	2/9/26 6:51:42.355 AM	2026-02-09T06:51:42.355329+00:00 ubuntu passwd[769]: password for 'root' changed by 'root'
2	2/9/26 6:51:42.355 AM	2026-02-09T06:51:42.355332+00:00 ubuntu groupadd[776]: group added to /etc/group: name=messi, GID=1000
3	2/9/26 6:51:42.355 AM	2026-02-09T06:51:42.355333+00:00 ubuntu groupadd[776]: group added to /etc/passwd: name=messi
4	2/9/26 6:51:42.355 AM	2026-02-09T06:51:42.355335+00:00 ubuntu groupadd[776]: new group: name=messi, GID=1000
5	2/9/26 6:51:42.355 AM	2026-02-09T06:51:42.355336+00:00 ubuntu useradd[782]: new user: name=messi, UID=1000, home=/home/messi, shell=/bin/bash, fromnone
6	2/9/26 6:51:42.355 AM	2026-02-09T06:51:42.355338+00:00 ubuntu useradd[782]: add 'messi' to group 'sudo'
7	2/9/26 6:51:42.355 AM	2026-02-09T06:51:42.355344+00:00 ubuntu useradd[782]: add 'messi' to shadow group 'sudo'
8	2/9/26 6:51:42.355 AM	2026-02-09T06:51:42.355345+00:00 ubuntu passwd[796]: password for 'messi' changed by 'root'
9	2/9/26	5036-02-09T06:51:42 1056249+00:00 ubuntu cvefeed-torinfo[8187]: New host: messiB

Add Data - Input Settings

Not Secure http://192.168.200.33:8000/en-US/manager/search/adddatamethods/inputsettings

Administrator Messages Settings Activity Help Find

Add Data Select Source Set Source Type Input Settings Review Done Back Review

Input Settings

Optionally set additional input parameters for this data input as follows:

App context

Application contexts are folders within a Splunk platform instance that contain configurations for a specific use case or domain of data. App contexts improve manageability of input and source type definitions. The Splunk platform loads all app contexts based on precedence rules. [Learn More](#)

App Context Search & Reporting [search]

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

Constant value

Regular expression on path

Segment in path

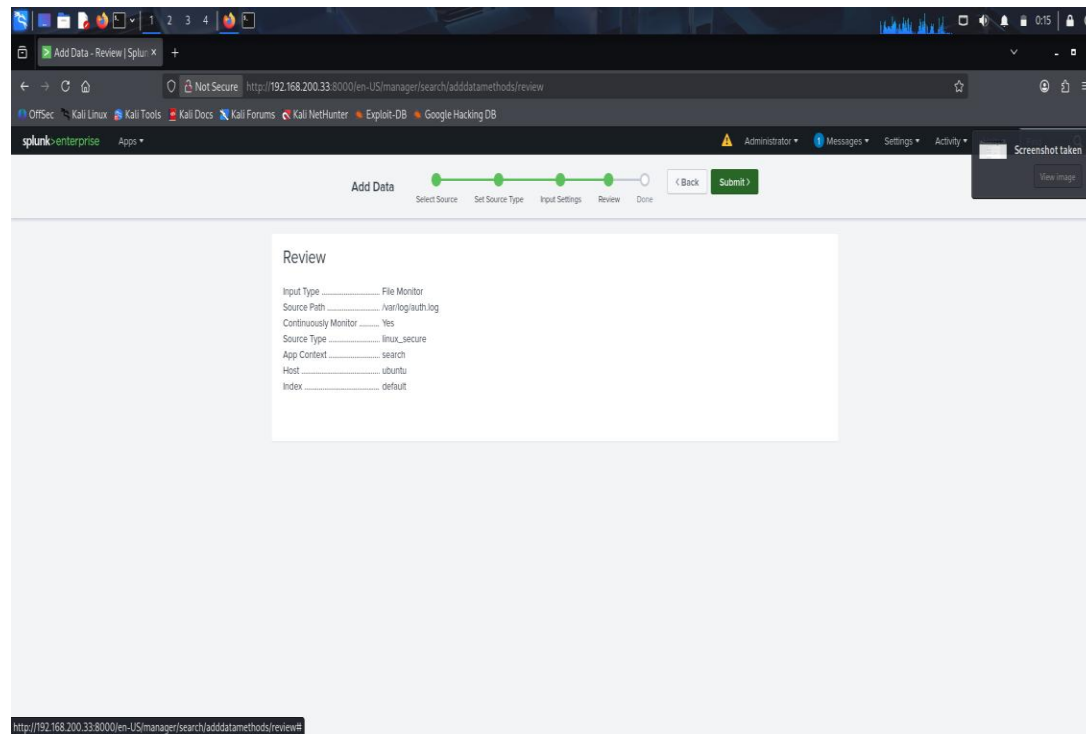
Host field value ubuntu

Index

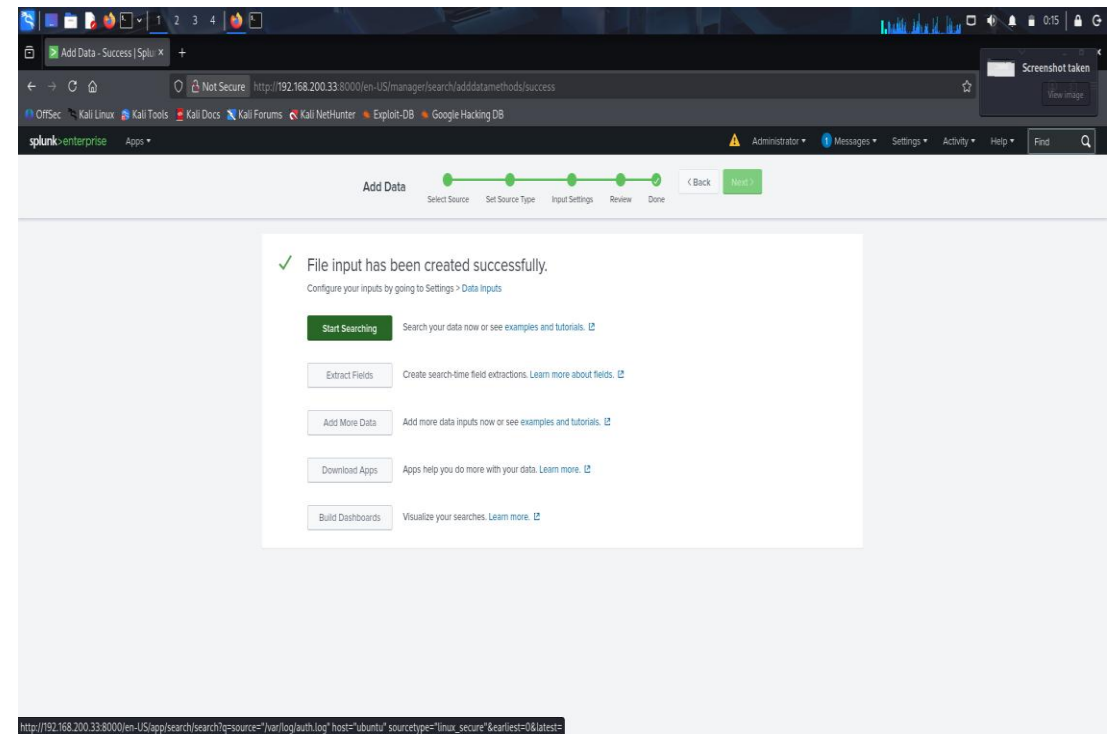
The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index Default Create a new index

Click – Submit



Click – Start searching



Search for failed SSH Login attempts

Search | Splunk 10.2.0

Not Secure http://192.168.200.33:8000/en-US/app/search/search?q=search source%3D"%2Fvar%2Flog%2Fauth.log" host%3D"ubuntu" sourcetype%3D"linux_secure"&earliest=0&latest=&sid

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

splunk>enterprise Apps Administrator 1 Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Modules Search & Reporting

New Search

source="/var/log/auth.log" host="ubuntu" sourcetype="linux_secure"

321 events (before 2/13/26 5:15:57.000 AM) No Event Sampling

Time range: All time

Save As Create Table View Close

Job

Splunk AI Assistant for SPL is now available.

Timeline format Zoom Out Zoom to Selection Deselect 1 hour per column

Format Show: 20 Per Page View: List

Hide Fields All Fields

SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

INTERESTING FIELDS
date_hour 4
date_mday 4
date_minute 42
a date_month 1
date_second 52
a date_wday 4
date_year 1
date_zone 1
a index 1
linecount 1

	Time	Event
>	2/13/26 5:15:01.721 AM	2026-02-13T05:15:01.721006+00:00 ubuntu CRON[3104]: pam_unix(cron:session): session closed for user root host = ubuntu source = /var/log/auth.log sourcetype = linux_secure
>	2/13/26 5:15:01.621 AM	2026-02-13T05:15:01.621203+00:00 ubuntu CRON[3104]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0) host = ubuntu source = /var/log/auth.log sourcetype = linux_secure
>	2/13/26 5:10:17.711 AM	2026-02-13T05:10:17.711943+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root host = ubuntu source = /var/log/auth.log sourcetype = linux_secure
>	2/13/26 5:10:12.390 AM	2026-02-13T05:10:12.390406+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by messi(uid=1000) host = ubuntu source = /var/log/auth.log sourcetype = linux_secure
>	2/13/26 5:10:12.386 AM	2026-02-13T05:10:12.386107+00:00 ubuntu sudo: messi : TTY=pts/2 ; PWD=/home/messi ; USER=root ; COMMAND=/opt/splunk/bin/splunk start --accept-license --answer-yes --run-as-root host = ubuntu source = /var/log/auth.log sourcetype = linux_secure
>	2/13/26 5:09:54.968 AM	2026-02-13T05:09:54.968915+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root host = ubuntu source = /var/log/auth.log sourcetype = linux_secure

Go to new search : Search - Failed password

After search failed password in splunk , it will display all recorded events where incorrect password attempts were made .

The screenshot shows the Splunk Enterprise web interface. At the top, there's a navigation bar with 'splunk>enterprise' and various tabs like Search, Analytics, Datasets, Reports, Alerts, Dashboards, and Modules. A search bar is present with the query: `source="/var/log/auth.log" "Failed password"`. Below the search bar, it indicates 321 events found. The results are displayed in a table with columns for Time and Event. The table shows 5 events related to failed password attempts and session management.

Time	Event
2/13/26 5:15:01.721 AM	2026-02-13T05:15:01.721006+00:00 ubuntu CRON[3104]: pam_unix(cron:session): session closed for user root host = ubuntu : source = /var/log/auth.log : sourcetype = linux_secure
2/13/26 5:15:01.621 AM	2026-02-13T05:15:01.621203+00:00 ubuntu CRON[3104]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0) host = ubuntu : source = /var/log/auth.log : sourcetype = linux_secure
2/13/26 5:10:17.711 AM	2026-02-13T05:10:17.711943+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root host = ubuntu : source = /var/log/auth.log : sourcetype = linux_secure
2/13/26 5:10:12.390 AM	2026-02-13T05:10:12.390406+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by messi(uid=1000) host = ubuntu : source = /var/log/auth.log : sourcetype = linux_secure
2/13/26 5:10:12.386 AM	2026-02-13T05:10:12.386107+00:00 ubuntu sudo: messi : TTY=pts/2 ; PWD=/home/messi ; USER=root ; COMMAND=/opt/splunk/bin/splunk start --accept-license --answer-yes --run-as-root host = ubuntu : source = /var/log/auth.log : sourcetype = linux_secure
2/13/26 5:09:54.968 AM	2026-02-13T05:09:54.968915+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root host = ubuntu : source = /var/log/auth.log : sourcetype = linux_secure

Step 4 : Failed Log Alert settings :

The screenshot displays the Splunk 10.2.0 web interface. On the left, the 'New Search' panel shows a search query: `source="/var/log/auth.log" "Failed password"`. Below the query, a table lists 6 events. The main panel on the right is titled 'Save As Alert' and contains the following configuration options:

- Title:** failed log in attempts
- Description:** Optional
- Permissions:** Private (selected) / Shared in App
- Alert type:** Scheduled (selected) / Real-time
- Expires:** 5 (selected) / minute(s)
- Trigger Conditions:**
 - Trigger alert when: Per-Result
 - Throttle: ☐
- Trigger Actions:**
 - + Add Actions
 - When triggered:
 - Severity: Medium

At the bottom of the 'Save As Alert' panel are 'Cancel' and 'Save' buttons. The background shows the Splunk search results interface with a table of events.

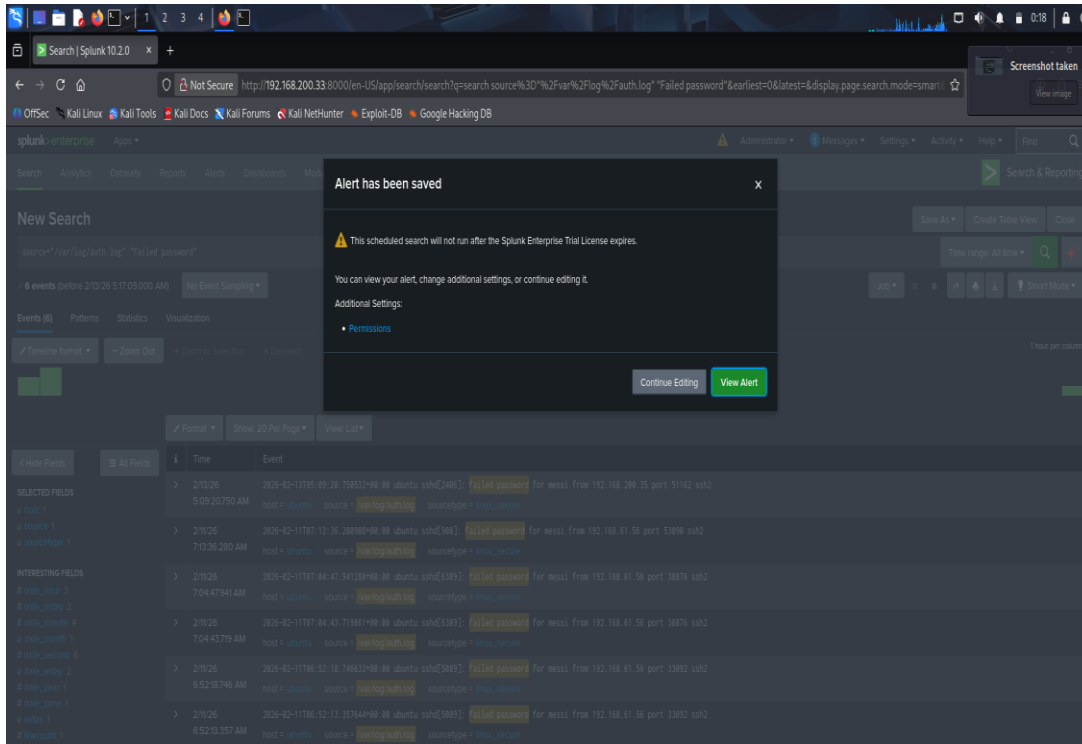
Time	Event
2/13/26 5:09:20.750 AM	2026-02-13T05:09:20.750Z host = ubuntu
2/11/26 7:13:36.280 AM	2026-02-11T07:13:36.280Z host = ubuntu
2/11/26 7:04:47.941 AM	2026-02-11T07:04:47.941Z host = ubuntu
2/11/26 7:04:43.719 AM	2026-02-11T07:04:43.719Z host = ubuntu
2/11/26 6:52:18.746 AM	2026-02-11T06:52:18.746Z host = ubuntu
2/11/26 6:52:13.357 AM	2026-02-11T06:52:13.357Z host = ubuntu source = /var/log/auth.log sourcetype = linux_secure

Set alert actions

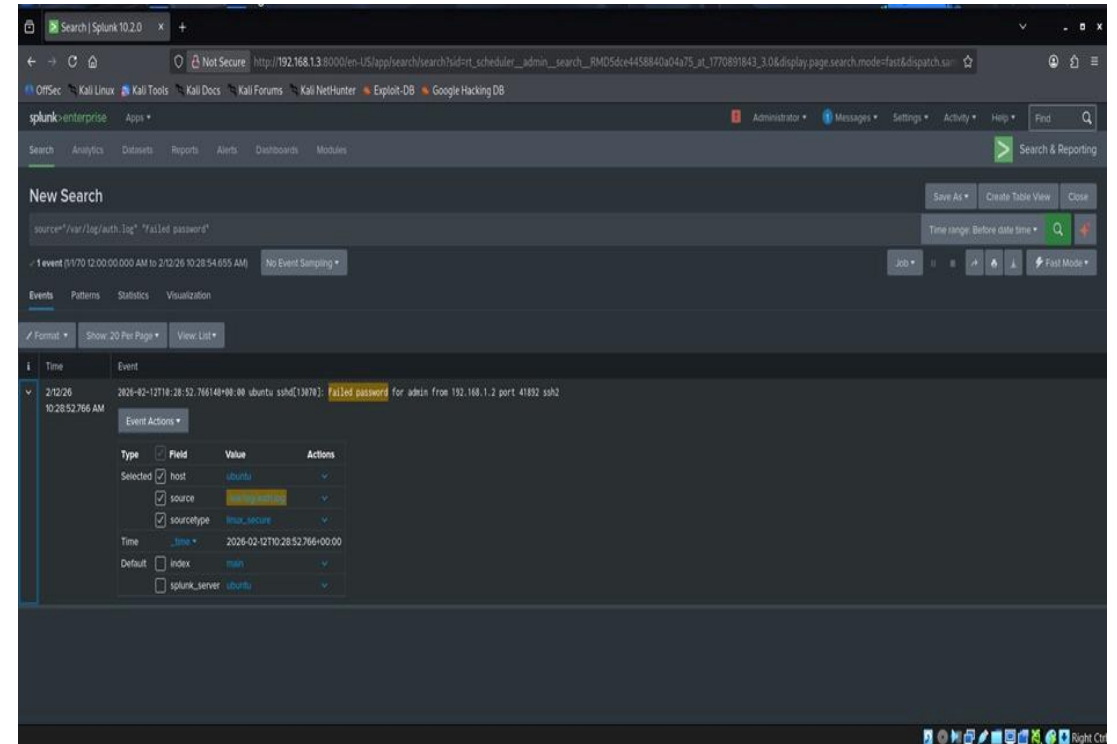
Title	: Failed login Attempts
Description	: Detection and alerting of unauthorized ssh login attempts using splunk
Alert type	: Real time
Expires	: 5 minutes
Ttrigger actions	: Add action – Select – Add to triggered alert
Security	: Medium

Click – Save

Alert has been saved : view alert



Failed login attempt



If you are trying to login with the wrong password in the ssh, it's shows alert in the splunk alert page , so we can track unauthorized SSH login attempts

Conclusion :

This project successfully demonstrated how Splunk can be configured to monitor and detect failed SSH login attempts by ingesting Ubuntu's `/var/log/auth.log` file and creating a real-time alert for "Failed password" events. By setting up log monitoring and alert mechanisms, we ensured that unauthorized access attempts are identified immediately, enabling quick response and reducing the risk of security breaches. Overall, this implementation highlights the importance of proactive system monitoring and shows how Splunk can effectively enhance security by detecting, tracking, and preventing unauthorized SSH access in a Linux environment.

Thank You !